# Introduction & Preliminaries

## 1. Introduction: what is Data Science?

Data science, also known as data-driven science, is an interdisciplinary field about scientific methods, processes, and systems to extract knowledge or insights from data in various forms, either structured or unstructured.[1]

Data science is a 'concept to unify statistics, data analysis and their related methods' in order to 'understand and analyse actual phenomena' with data. It employs techniques and theories drawn from many fields within the broad areas of mathematics, statistics, information science, and computer science, in particular from the subdomains of machine learning, classification, cluster analysis, data mining, databases, and visualisation.[2]

The ability to take data, to be able to understand it, to process it, to extract value from it, to visualize it, to communicate it, is going to be a hugely important skill in the next decades, not only at the professional level but even at the educational level for elementary school kids, for high school kids, for college kids. Because now we really do have essentially free and ubiquitous data. So the complimentary scarce factor is the ability to understand that data and extract value from it.[3]




**Figure 0.1.** There's a lot of hype around Data Science.

[1] Source: `https://en.wikipedia.org/wiki/Data_science`
[2] See footnote 1.
[3] Source: 'Hal Varian on how the Web challenges managers', McKinsey & Company

There *is* something new in "data science". It's not a science. It's more of a process; a merging of firstly, skills in applied mathematics, computer science, statistics, visualisation and communication, with secondly, rich data sets and domain knowledge. In practice, it's usually done in teams. No one has deep knowledge of statistics, mathematics, computer science, visualisation together with detailed knowledge of the data source, industry as well as its interesting questions.

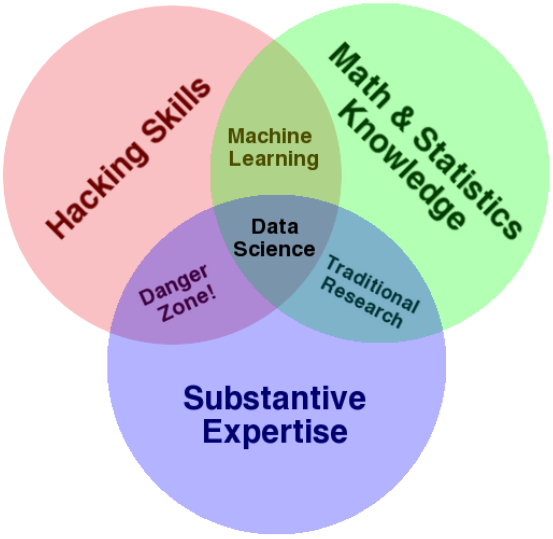



**Figure 0.2.** Venn diagram: Where does data science fit?

There's no academic field called 'data science'. After all, scientists have always used data. Statisticians theorise about data, and analyse it too. Very few academics are 'professors of data science'. Cathy O'Neil's book 'Doing Data Science' says that an academic data scientist might be defined as: 'a scientist, trained in anything from social science to biology, who works with large amounts of data, and must grapple with computational problems posed by the structure, size, messiness, and the complexity and nature of the data, while simultaneously solving a real-world problem.'[4]

There are a lot of jobs as data scientists. What do these people do? Fundamentally, a data scientist is someone who can extract meaning from data and communicate the results clearly. Data science at a company typically includes:

- data collection, storage and management
- who has access to what data
- how will the data be used, how does it add value
- privacy considerations
- analysis of data
- communicating the results

## 2. Case studies: successes of data science

Moneyball! Data in baseball outperforms standard predictors of who will be successful. There's a book, and a movie with Brad Pitt. Other big success areas are:

- Spam filters

[4] O'Neil, Cathy, and Rachel Schutt. *Doing data science: Straight talk from the frontline.* 'O'Reilly Media, Inc.', 2013.

- Fraud detection
- Election predictions (Nate Silver, `https://fivethirtyeight.com`)
- Predictions of crime hotspots
- Text analysis: twitter, blogs
- Targeted advertising
- Song, movie and product recommendations
- Forecasting (e.g. energy demand prediction, finance)
- Imputing missing data (e.g. netflix recommendations)
- Detecting anomalies (e.g. security, fraud, virus mutations)
- Classifying (e.g. credit risk assessment, cancer diagnosis)
- Ranking (e.g. Google search, personalization)
- Summarizing (e.g. news zeitgeist, social media sentiment)
- Decision making (e.g. AI, robotics, compiler tuning, trading)

Examples of specific questions that one could ask:

- Predict whether a heart attack patient will have a second heart attack, using demographic, diet and clinical data.
- Predict the price of a stock in 6 months, using company performance measures and economic data (but don't invest your money by your answers...).
- Identify the numbers in a handwritten ZIP code, from a digital image.
- Estimate the amount of glucose in the blood of a diabetic person from the infrared absorption spectrum of that persons blood.
- Identify the risk factors for prostate cancer, based on clinical and demographic data.
- Predict which flu strain will be successful next year based on its DNA sequence and relationships to other flu strains.

## 3. The process of data science

In most cases, doing data science will consist of the following pipeline:

- The science: what's a good question and what's the data set you will use to answer it?
- What is the input and what are you trying to determine?
- Data handling, cleaning and exploration: using computer programming and as well as knowing things
- The maths and the main analysis:
  - statistics, computing: what do you compute, why, and what does it mean?
  - machine learning: supervised, unsupervised..
  - networks: characteristics, analysis, comparisons
- Communication: words, visualisations, summaries to tell the story

In this course, you will be exposed mainly to the mathematical part of this workflow, although some information on the other aspects here will be mentioned in passing, too.

## 4. Aims and outcomes of this course

As mentioned above, this course's main aim is to equip you with the necessary skills to understand and work with the mathematical side of data science. To this end, you will learn about:

- The mathematicalconcepts underpinning methods in learning from data
- The process of conceptualisation of the analysis
- The process of explanation of the analysis
- The mathematical justification of the analysis
- Getting a good exposure to current methods and their use in practice in dealing with (realistic) data
- What the tools are, how they work mathematically, and how to analyse a data set and clearly communicate the results

## 5. Exploratory data analysis

In the pipeline described previously, one of the very first steps after determining the main question and data necessary to answer it, is conducting some *exploratory data analysis* (EDA). Whilst this course will not go in-depth on how to do this, we will give an overview of what it entails.

You will not have perfect data. The goal is to transform this into something tractable. This is why EDA is needed, it helps you to:

- Find mistakes! (negative heights? copies of columns? unreasonable values? missing values?)
- Get an idea of whether the signal you want is actually in your data (if so, where? Maybe your problem is simple, so just one predictor works. Find the direction and size of relationships between predictors and outcome.)
- Check the existence of statistical relationships between predictors
- Select your methods and metrics based on the type of task and analysis you want to carry out

Some data will be rubbish. EDA is important to sift out the good from the bad!

## 6. Some preliminary definitions

We now give some definitions of terms used throughout the course.

**6.1. The different types of data.** Data in its very simplest form usually consists of a big spreadsheet, a table, or a data frame full of numbers and categories. Typically, there is one row per observation, and one column per variable, i.e. all the predictors and the outcome(s). Reading 37 columns and 117,000 rows of a spreadsheet is not helpful. We need to summarise and visualise the data set in lots of ways to explore it, choose our method, find mistakes, etc. Broadly speaking, there are two classes of data, that you will find:

6.1.1. *Categorical data.* A *categorical* (or *discrete*) variable takes on one of a limited number of values (usually fixed), such as classes, or memberships. This means that all observations in the data will fall into $k$ classes, for example. In essence, this leads to a discrete space of values. Some examples include: team name, which county someone lives in, breed of a dog, modules (mathematics, physics, computer science). The distribution is really just the counts or frequencies of the different values. Predicting a categorical variable is called *classification*: we want to classify a new point into the right category.

6.1.2. *Quantitative data.* A *quantitative* (or *continuous*) variable is numerical, and represents a measurable quantity. This is usually defined on a continuous space. Some examples are: height, salary, stock price, profit, speed. In some cases, it's important that your data is ordered and can be ranked. This means that you have *ordinal* data, e.g. number of bedrooms, number of players, number of people in a city (nearly continuous for practical purposes). The distribution of quantitative data usually has a mean, variance, skewness etc. Predicting a quantitative variable is called *regression*.

6.1.3. *Other, more specific types of data.* Note that aside from the two main categories above, there exist some more data types that are less common.

For example, *time series* data, which can be seen as a special case of ordinal quantitative data, as is usually also numerical. However, in this case, the order usually follows real-life timestamps, where data is collected at regular (or irregular) intervals over some amount of time. Some examples are: average number of home sales for many years, stock prices.

Sometimes you will also encounter simple *text* data. In this case the data contains words, sentences and sometimes even whole books. Usually, the first step will be to convert this data into something more tangible for the computer, i.e. numbers. Examples can be found in modern-day translation (such as Google Translate) and more generally, the field of *Natural Language Processing*.

**6.2. Predictors vs outcomes.** The first step to exploring the data is to determine and declare which variables are *predictors* (sometimes also known as *descriptors*) and which are *outcomes*. These can sometimes also be called *input/output*, respectively. The game of this course is to establish relationships between these two spaces: be aware that this is a decision that you make.

6.2.1. *Predictor variables.* Examples of predictor variables are:

- numbers
- continuous measurements like height, price, profit, concentration
- discrete measurements (integers): number of people, counts
- text: tweets, emails, patient records
- images: handwriting, medical images, photos
- even DNA sequences..
- combinations of these things

Be aware that randomness happens! It happens both in the process of getting the data, and in the form of noise in the actual observations. Simply declaring certain variables of data as predictors should not rule out the possibility that they are noisy, much like the outcome variables.

6.2.2. *Outcome variables.* This is the piece of the data set that we will aim to model and predict. What might your outcome (or "label") variables look like?

- Continuous data: in this case the problem is *regression*.

- Categorical data: in this case the problem is *classification*.

**6.3. Learning from data.** Inputs belong to an input space: $x^{(i)} \in X$. For example, if you have $p$ different continuous features (height, weight, grade on exam, peak exhaled air flow, blood hemoglobin...) for each data point, then $x^{(i)} \in \mathbb{R}^p$. In this case, our data is *multivariate*. $x^{(i)}$ could also contain some other data that are not real numbers (i.e., categorical): county of residence, gender, company name...

Outputs $y^{(i)}$ as well can be continuous or categorical: some number or feature or group, some knowledge about the data point. Fundamentally, supervised machine learning is about finding ways to link the predictors $x^{(i)}$ to the outcomes $y^{(i)}$ so that we can make new predictions, i.e. generalise to unseen data. Indeed, there are two big, main classes of problems/algorithms:

6.3.1. *Supervised learning.* This is the case where there is a known outcome variable: the truth. For example:

- emails where we we know which ones are spam
- patients for whom we know their eventual outcome (heart attack or not)
- patients where we know their glucose level
- stocks and their prices 6 months later
- tweets and their author (election team or Donald himself)

We want to be able to predict that outcome for new observations of the input (predictor) variables.

The first part of this course will exclusively consist of methods that perform supervised learning. The latter few chapters will deal with it's counterpart:

6.3.2. *Unsupervised learning.* In this case, there is not a known outcome variable that we want to predict. Instead, we want to understand how the data are intrinsically organised, clustered, related. This is something that we want to extract from the data. For example:

- use diverse measurements (blood, tumour shapes, gene expression) to identify similar types of breast cancers
- explore differences in the bacterial composition inside the guts of healthy vs unhealthy individuals
- gain intuition for very high-dimensional data; make it more tractable